

Please solve the following exercises at home prior to the tutorial:

Exercise 1

Given is the alphabet $A = \{a, e, i, o, u\}$ with occurrence probabilities
 $p(a) = 0.25$, $p(e) = 0.2$, $p(i) = 0.1$, $p(o) = 0.3$, and $p(u) = 0.15$.

Calculate the information contents of the symbols in A and the entropy of A.

Exercise 2

Provide simple logical operations using bit masks* for the following manipulations of ASCII characters:

- Extraction of the numerical value 0 to 9 from the ASCII-Codes for the characters 0 to 9.
- Conversion from lower case to upper case and vice versa.

* A bit mask is a series of binary digits which – using certain logical operations (e.g., AND, OR, NOT, XOR, ...) on the ASCII-Code – yield the desired result.

The following exercises will be done during the tutorial:

Exercise 3

Given are the following two binary codes for the alphabets $A = \{x_i\}$ and $B = \{y_i\}$ as well as the respective occurrence probabilities p_i :

x_i	a	e	i	o
p_i	0.3	0.25	0.3	0.15
Code	10	010	00	0110

y_i	1	2	3	4
p_i	0.7	0.05	0.1	0.15
Code	0	1000	111	110

- Considering a typical message of length n , what is the size in bits of a binary coded message consisting of symbols from either alphabet A or alphabet B?
- By how many percent is the longer message longer than the shorter one?
- Determine the entropy of the alphabets A and B.
- How large is the redundancy of a typical message from A or B?